O S O S

G T G T

5 E 5 E

N O N O

O S O S

G T G T

5 E 5 E

N O N O

O S O S

G T G T

5 E 5 E

Ex 2.1

$\varepsilon = 0.5$

$a_1 \qquad a_2$

$$1 - \varepsilon + \frac{\varepsilon}{2} = 0.5 + \frac{0.5}{2} = \underline{\underline{0.75}}$$

Ex 2.2

	1	2	3	4
θ_1	0	0	0	0
θ_2	-1	0	0	0
θ_3	-1	1	0	0
θ_4	-1	$\frac{1-2}{2} = \frac{-1}{2}$	0	0
θ_5	-1	$\frac{1-2+2}{3} = \frac{1}{3}$	0	0
θ_6	-1	$1/3$	$0/1$	0

$\theta_2 - \theta_6 \rightarrow$ was ε- greedy selection

Ex 2.3

The $\varepsilon = 0.01$ will do better because it has a higher slope. In the longer run as the estimates converge the significance of 'ε' will reduce, causing the random action selection to be mostly sub-optimal.

It will be better by 9% ??

Ex 2.4

$$\theta_{n+1} = \left(1-\frac{1}{n}\right)^n \theta_1 + \sum_{i=1}^{n} \frac{1}{n}\left(1-\frac{1}{n}\right)^{n-i} R_i \quad ??$$

Ex 2.6

We see the oscillations in the early part of the optimistic initial value algorithm because due to disappointing itself after pulling a lever, the agent thinks that the next lever is the optimal one.

Ex 2.7 ?

Ex 2.8

In the 11TH iteration after all